

Examance

Privacy-first, zero-knowledge-encrypted anonymous exam grading

No account required · Encrypted at rest · GDPR by design

The problem in the teachers' lounge

The workflow	The bias (halo effect)	Data protection
Manual correction, transcribing points into spreadsheets, and computing statistics eat into teachers' limited time.	A name on the cover sheet unconsciously influences grading before a single answer is read.	Mainstream cloud edtech tools are frequently a poor — sometimes illegal — fit for sensitive pupil data under GDPR/BDSG.

Examance addresses all three at once: automates the mechanical parts of grading, structurally removes the student's identity from the corrector's view, and — by default — never sends sensitive data to a server at all.

Getting started — no account required

Two entry points, changeable later in Settings: a fully local, offline-first workspace with **zero server dependency**, or a school-managed cloud account for cross-device sync. Either way, data is encrypted client-side (Argon2id → HKDF-SHA-256 → AES-256-GCM) before it is ever written to disk.



Welcome to Examance

Privacy-First Anonymous Exam Management & Grading



No Account Required

Start Local Workspace

Ideal for solo offline grading. All data is encrypted with local keys and stored directly in your browser.

- ✨ No registration needed
- 🔒 Encrypted at rest with your passphrase
- 📁 Export/Import workspace as .bgproj file

Choose a passphrase

At least 12 characters

Repeat passphrase



School Account

Connect to Cloud Server

Sync exams across devices and connect with your institution's backend server.

Backend Server URL

e.g. `http://localhost:8000`

Email

teacher@school.example

Password

Enter password

Your exams at a glance

 **Examance**

DashboardExercise LibraryAnalyticsSettings

WorkspaceLocal ModeConnect to CloudLock

Exams Dashboard

Manage, grade, and analyze your Schulaufgabe exams

Total Exams
0

Configured Subjects
0

Grade Levels
0

Global Analytics
[View Multi-Exam Stats →](#)



Welcome to Examance!

You don't have any exams in your workspace yet. Follow these quick steps to get started:

1

Create an Exam

Configure LaTeX template, total points, and grade thresholds.

2

Link Exercises

Add questions from your Exercise Library or create new ones.

3

Scan & Grade

Upload student PDFs, anonymously score, and review analytics.

+ Create First Exam

Or Import .bgproj Archive

Import .bgproj

Create New Exam

Impressum · Datenschutz

All Local | LaTeX Local

No Server Configured v0.0.0-dev

4

Pillar 1 — Reusable exercise library & LaTeX authoring

Exercises live in a **shared, taggable library** (topic, grade, subject) instead of being copy-pasted between exam files each term. Each item is a LaTeX fragment with live PDF preview, automatic point-value parsing, and free-text / single-choice / multiple-choice support.

The screenshot shows a 'Create New Exercise' dialog box with the following fields and options:

- Name ***: For-Schleifen zaehlen
- Topic ***: Schleifen
- Grade**: 9
- Subject**: Informatik
- Variant Key**: e.g. Moebel
- Exercise Type**: ☒ Free Text ☐ Multiple Choice (MC)

Below the form, there are two tabs: 'LaTeX Source Code' (selected) and 'Live Preview PDF'. The 'LaTeX Source Code' tab shows a preview of the composed LaTeX source code:

```
1 Wie oft wird der Inhalt einer For-Schleife mit dem folgenden Kopf
2 ausgefuehrt?
3 \begin{verbatim}
4 for (int i = 1; i < 10; i += 2)
5 \end{verbatim}
```

The 'Live Preview PDF' tab is currently empty, displaying the text 'Click 'Live Preview PDF' to render'. On the right side of the dialog, there is a vertical sidebar with a 'Solution PDF' icon and a 'Solution PDF' label.

Pillar 1 — Variants, versions, and exam assembly

The library groups exercises, tracks **variants** (anti-copying A/B/C phrasings of the same problem) and **versions** (fix history with diffing) — filterable by grade, subject, topic.

The screenshot shows the Examance Exercise Library interface. At the top, there's a navigation bar with the Examance logo, links to Dashboard, Exercise Library, Analytics, and Settings, and buttons for Workspace, Local Mode, Connect to Cloud, and Lock. Below the navigation bar, the main heading is "Exercise Library (Aufgabenkatalog)" with a subtitle "Reusable LaTeX exercise collection live-linked across your exams." and a "+ Create New Exercise" button. On the left, there's a search bar "Search exercises by nar" and filters for Grade (All Grades) and Subject (All Subjects). Below these are topic buttons: "All Topics (2)", "Schleifen (1)", and "Vererbung (1)". The main content area displays two exercise cards. The first card is titled "For-Schleifen zaehlen" and includes tags for Schleifen, Klasse 9, Informatik, 0 Pkt, and 1 variant. The second card is titled "Vererbung Grundlagen" and includes tags for Vererbung, Klasse 10, Informatik, 0 Pkt, and 1 variant. Both cards have a "_General v1" dropdown menu.

Examance

Dashboard Exercise Library Analytics Settings

Workspace Local Mode Connect to Cloud Lock

Exercise Library (Aufgabenkatalog)

Reusable LaTeX exercise collection live-linked across your exams.

+ Create New Exercise

Search exercises by nar

Grade: All Grades

Subject: All Subjects

All Topics (2)

Schleifen (1)

Vererbung (1)

For-Schleifen zaehlen Schleifen Klasse 9 Informatik 0 Pkt 1 variant

_General v1

Vererbung Grundlagen Vererbung Klasse 10 Informatik 0 Pkt 1 variant

_General v1

Pillar 1 — Assembling the exam

Pick exercises from the library, configure the grading key (linear, *Oberstufe*-weighted, or custom cutoffs 1–6), and Examance renders a print-ready, QR-coded booklet locally via WASM XeLaTeX (Tectonic) — no source ever leaves the browser in local mode.

 **Examance**

DashboardExercise LibraryAnalyticsSettings

WorkspaceLocal ModeConnect to CloudLock

Create Exam (Assembly from Exercise Library)

1. Exam Metadata

Exam Title

e.g. 2. Kurzarbeit 10a Informatik

Testart (\Testart)	Jahrgangsstufe (Grade)	Klasse / Kurs	Nummer (\Nr)
Kurzarbeit	10	a	1

Vorschau `\Klasse{10a}` : 10a

Datum (\Datum)	Dauer	Fach (\Fach)	Lehrer Nachname (\Lehrernachname)
17.8.2026	30 Minuten	Informatik	Her

Header Info Instructions (\Info)

```
\begin{itemize}
\item Die Arbeit wird anonymisiert korrigiert. Trage deine Initialen ins QR-Code-Feld ein.
\item Mit Bleistift oder rotem / blauem Korrektiergerät (nicht für die Korrektur geeignet) zu verwenden.
```

 **Notenschlüssel (Noten 1 bis 6)**

Klassisch (50% = Note 4)Oberstufe (40% = Note 4)Gleichmäßig

Pillar 2 — Digital, anonymous correction

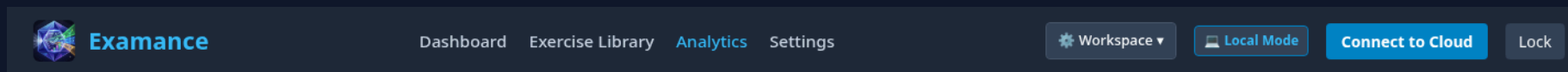
1. **Write** — students complete the printed, QR-coded exam with pen and paper.
2. **Scan** — the stack goes through the school scanner into a single PDF.
3. **Split & encrypt** — Examance splits by QR code and encrypts every page client-side.
4. **Grade** — on-screen pen/mouse annotation over the scan; originals stay untouched.

Free-text: teacher sees handwriting and answer — **not** the student's name.

MC / SC: WebAssembly optical-mark-recognition (OMR) auto-detects marks and applies penalty logic — no manual tallying.

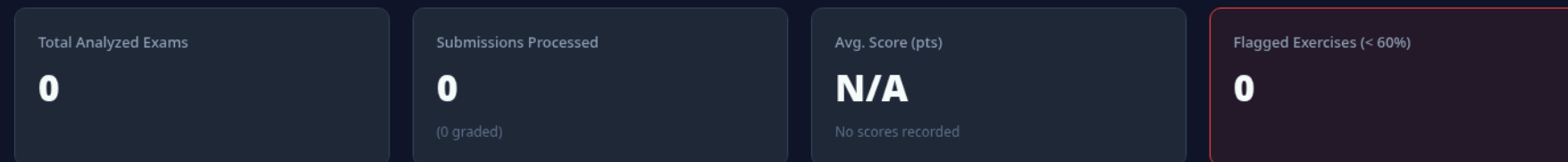
Pillar 3 — Didactic analytics, privacy-preserving

Score distributions, per-topic heatmaps for knowledge gaps, exercise/question quality metrics across years, variant-fairness comparisons, and CSV/XLSX export — all computed from the encrypted local vault, the same data that was never uploaded anywhere.



Global Multi-Exam Analytics

Cross-exam performance tracking, exercise quality metrics & variant comparison



No graded submissions recorded yet. As student scans are uploaded and scored, real-time cross-exam averages, quality metrics, and variant difficulty comparisons will automatically calculate below.

Exercise Variant Difficulty & Fairness Comparison

Compare performance between different question variants (e.g. Gruppe A vs Gruppe B) to detect unintended difficulty imbalances.

Exercise & Question Quality Metrics

Identify questions that consistently produce low average scores across multiple years or exam sessions.

ies, and results are stored:

scans stored 100% locally in your browser IndexedDB.

BlindGrade server.

ts Local)

out student identities and grade submissions stay 100% on your local device.

t of storage strategy):

source to any server.

uires authenticated account.

No compromises on data protection

- **All Local** (*default*) — encrypted entirely in the browser (IndexedDB), zero bytes reach a server. Export/import as a password-protected `.bgproj` archive.
- **All Server** — synced as **AES-256-GCM ciphertext**; the server never sees plaintext.
- **Hybrid** — shared exercise library on the server, identities & results stay local.

LaTeX compiles locally (WASM, nothing leaves the browser) or server-side. The local engine auto-routes around hardware limits: sequential rendering on low-spec/eco devices, multi-core/SIMD on capable machines.

Conclusion — value at every level

For teachers	For students	For school leadership
Real time savings: automatic MC/SC scoring, reusable library, no manual grade tallying.	Guaranteed fairness — objective, anonymous grading via identity/answer decoupling.	Legal certainty: privacy-by-design, no compromise on GDPR / Bavarian school-law protection.

Examance — Privacy-First Anonymous Exam Management